

**DATA ANALYTICS WITH POWER BI
PROJECT REPORT**

(Project Semester August-December 2025)

Crime Insights & Intelligence Portal

Submitted by
Deepak

Registration No. 12316722

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Under the Guidance of

Vikas Mangotra

UID: 31488

Assistant professor

Discipline of CSE/IT

Lovely School of Computer science and Engineering

Lovely Professional University, Phagwara

DECLARATION

I, **Deepak** student of **Computer science and Engineering** under CSE/IT Discipline at, Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date: 12-12-2025

Signature:

A small rectangular box containing a handwritten signature in blue ink. The signature appears to be 'Deepak' written in a cursive style.

Registration No.: 12316722

Name of the student: Deepak

CERTIFICATE

This is to certify that Deepak bearing Registration no. 12316722 has completed INT374 project titled, “Crime Insights & Intelligence Portal” under my guidance and supervision. To the best of my knowledge, the present work is the result of his original development, effort and study.

Signature and Name of the Supervisor

Designation of the Supervisor

School of Computer science and Engineering

Lovely Professional University

Phagwara, Punjab.

Date: 12-12-2025

Acknowledgement

I would like to express my sincere gratitude to my faculty, Vikas Mangotra, for her continuous guidance, valuable feedback, and consistent support throughout this project.

I would also like to thank Lovely Professional University for providing the resources and academic environment that allowed me to complete this project successfully.

Last but not least, I extend my thanks to my friends for their encouragement and support during the course of this work.

Deepak

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1. Introduction

In today's data-driven world, understanding crime patterns is essential for improving public safety, operational planning, and strategic decision-making. With the increasing volume of crime data generated every year, traditional analysis methods are no longer sufficient. This led to the development of the Crime Insights & Intelligence Portal, an interactive analytical system designed to uncover hidden patterns, trends, and hotspots within crime records.

This portal leverages modern data visualization techniques to provide a clear, intuitive, and comprehensive view of crime statistics from 2020 to the present. By integrating key metrics—such as crime frequency, victim demographics, weapon usage, district-level crime intensity, and hotspot mapping—the system empowers law-enforcement agencies, policymakers, and analysts to make informed, evidence-based decisions.

The goal of this project is to transform raw crime data into meaningful insights that enhance situational awareness, support resource allocation, and strengthen crime-prevention strategies through a visually intelligent Power BI dashboard.

2. Source of Dataset

The dataset used in this project was sourced from the Catalog, an open platform that provides real-world datasets for data analysis and visualization practice.

Dataset Name: crime-data-from-2020-to-present

Source Link: <https://catalog.data.gov/dataset/crime-data-from-2020-to-present>

The dataset consists of several fields that capture detailed crime-related information from 2020 to the present, such as:

- **Crime Details:** Core crime-level information including crime type, date, time, area name, district number, modus operandi, and case status.
- **Victim Information:** Attributes related to victims such as age, sex, and descent, allowing demographic-based crime analysis.
- **Weapon Usage:** Data describing the type of weapon used (if any), ranging from firearms to strong-arm force and other weapon categories.
- **Location & Geography:** Geographic coordinates (latitude and longitude) enabling hotspot mapping and spatial crime clustering.
- **Premise & Environment:** Information on the type of location where the crime occurred, such as streets, residences, parking lots, or commercial areas.

This dataset provides an excellent foundation for exploring crime behaviour over time, analyzing emerging patterns, identifying high-risk areas, and building intelligent crime-visualization dashboards.

3. Dataset Preprocessing

Preprocessing is a crucial step in data analysis to ensure consistency, accuracy, and reliability.

The crime dataset (2020–Present) underwent the following preprocessing steps:

3.1 Handling Missing Values

The dataset contained missing or blank values in several key fields including:

- Victim Age
- Victim Descent
- Weapon Type
- Case Status
- Coordinates (Longitude/Latitude)

The following actions were taken:

- Missing categorical values were replaced with “**Unknown**”.
- Invalid or zero geographic coordinates were removed.
- Empty age fields were either excluded from age-related analysis or grouped as “**Not Available**”.

3.2 Data Cleaning

To make the dataset consistent and analysis-ready:

- Standardized date fields into **Year**, **Month**, and **Time** formats.
- Converted text columns (Crime Type, Area Name, Weapon Type) into uniform casing.
- Removed duplicate crime entries.
- Ensured numeric fields (e.g., victim age, district number) were stored in correct data types.
- Fixed inconsistencies in status descriptions (e.g., merging similar status labels).

3.3 Feature Engineering

New derived fields were created to enhance analytical insights:

- **Time Slot** (Morning, Afternoon, Evening, Night) based on crime time.
- **Crime Category**: Classified crimes into *Major* and *Minor* categories.
- **Age Group Bins** to visualize demographic distribution more effectively.
- **Weapon Frequency Categories** to identify commonly used weapons.
- **District Crime Intensity** for trend comparison across districts.

3.4 Merging Datasets

To enable dashboard interactivity:

- Separate tables were modeled using relationships in Power BI based on:
 - Crime Type

- Weapon Type
 - Area Name
 - Year
- Geographic fields were connected to map visuals for hotspot analysis.

These preprocessing steps ensured that the dataset was optimized for creating a comprehensive, interactive crime analytics dashboard.

Analysis on Dataset

4.1 Crime Trend Over Time

i. General Description

Crime trend analysis helps understand how crime has changed from 2020 to the present. Examining year-wise fluctuations reveals increases, declines, and potential influences such as population growth, societal events, or policing initiatives. This analysis supports strategic planning by highlighting periods when crime rates were highest or lowest.

ii. Specific Requirements

- Evaluate the total number of crimes reported each year.
- Identify years with upward or downward crime trends.
- Compare long-term stability versus sudden shifts.
- Understand whether crime frequency is rising or declining over time.

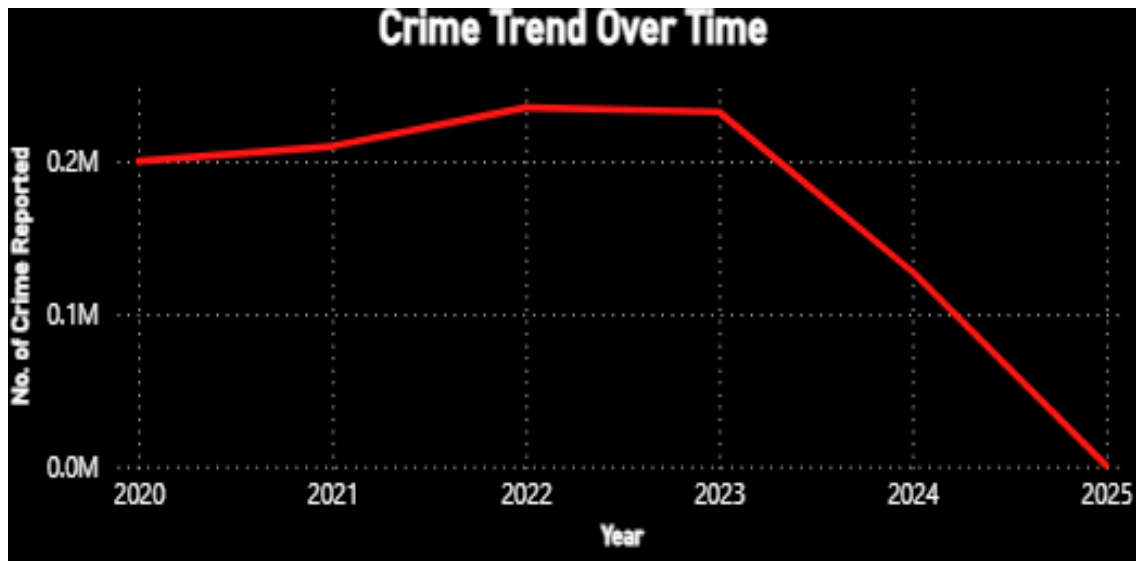
iii. Analysis Results

The trend chart shows a gradual rise in crime from **2020 to 2023**, with **2023 being the peak crime year**, crossing approximately **0.25M** reported cases. However, after 2023, crime volumes saw a sharp decline, especially in **2024 and 2025**, where reported cases dropped dramatically to near zero levels.

This significant reduction in 2024–2025 suggests either a major improvement in enforcement, changes in reporting policies, or incomplete datasets for recent years.

iv. Visualization Interpretation

The line graph clearly highlights crime escalation until 2023 followed by a sharp drop. This visualization emphasizes the need to investigate factors contributing to the sudden downturn in later years.



4.2 Top 7 Crime Types

i. General Description

Different crime categories occur at different frequencies, and analyzing these helps identify the most common offenses affecting the region. Understanding top crime types assists law enforcement in prioritizing resources and designing targeted intervention strategies.

ii. Specific Requirements

- Rank crime types based on total number of cases.
- Compare major vs minor crime prevalence.
- Identify crime categories that require the most attention.

iii. Analysis Results

The analysis shows that Vehicle Theft is the most frequently reported crime, exceeding 100K cases. It is followed by:

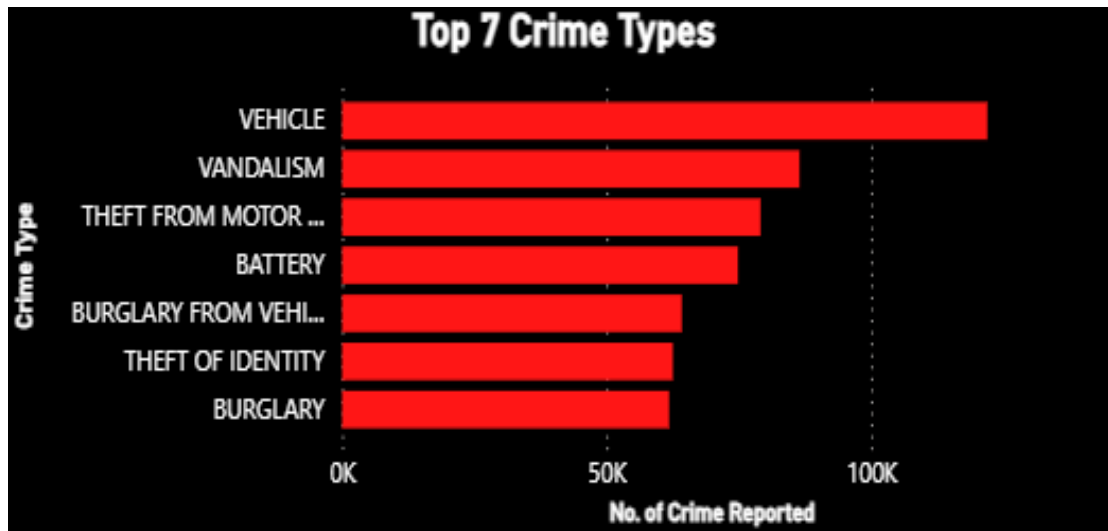
Vandalism

- Theft from Motor Vehicle
- Battery
- Burglary from Vehicle
- Theft of Identity

- Burglary
- Vehicle-related crimes dominate the dataset, indicating mobility-related vulnerabilities and urban property crime as major concerns.

iv. Visualization Interpretation

The bar chart visually emphasizes the disproportionately high volume of vehicle-related offenses. These crime types may require enhanced surveillance in high-traffic and parking areas.



4.3 Peak Crime Hours

i. General Description

Analyzing crime by hour reveals when criminal activity is most common. Time-based crime trends help allocate patrol resources effectively and identify high-risk periods throughout the day.

ii. Specific Requirements

- Categorize crime occurrences into Morning, Afternoon, Evening, and Night.
- Compare crime volumes across each time slot.
- Identify high-risk periods for both major and minor crimes.

iii. Analysis Results

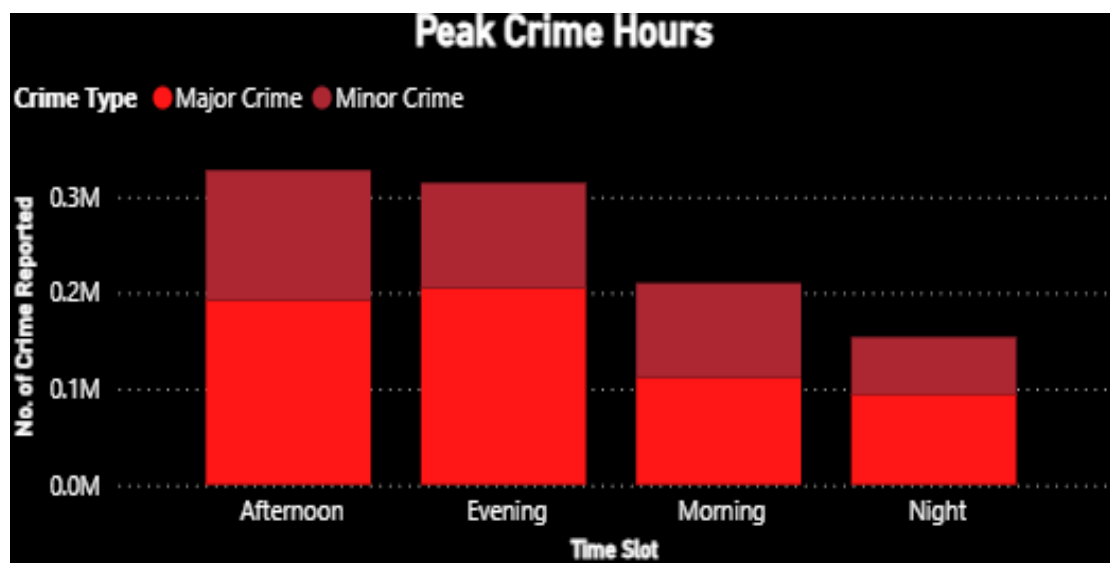
The highest crime frequencies occur during the **Afternoon** and **Evening**, with both major and minor crimes peaking in these time slots.

Morning and Night show lower crime volumes, indicating decreased activity during early and late periods.

This suggests that increased movement, workplace activities, and public interactions during afternoon and evening contribute significantly to crime exposure.

iv. Visualization Interpretation

The stacked bar chart shows clear dominance of crime activity in afternoon and evening hours. This information supports the need for enhanced patrol deployment during these periods.



4.4 Crime Hotspot Heatmap — Los Angeles

i. General Description

Hotspot analysis is a critical component of crime intelligence, as it identifies geographic regions where crime incidents are concentrated. Heatmaps use spatial data (latitude and longitude) to visualize crime density, helping law-enforcement agencies understand where crime is most intense and where patrol allocation or preventive measures should be prioritized.

The Los Angeles hotspot heatmap provides a high-level spatial overview of crime concentrations across the city, highlighting zones with increased criminal activity.

ii. Specific Requirements

- Use geographic coordinates to map crime density across Los Angeles.

- Identify high-risk clusters and densely affected neighborhoods.
- Compare hotspot intensities across central, coastal, and suburban areas.
- Support decision-making for patrol planning and resource allocation.

iii. Analysis Results

The hotspot map clearly reveals several concentrated crime clusters within the Los Angeles region:

- Downtown Los Angeles shows the highest density, indicated by deep red zones.
- Surrounding areas such as Hollywood, Westlake, South Central, and Mid-City also appear as significant hotspots.
- Regions around Long Beach and Inglewood show moderate to high activity.
- Suburban areas farther from central LA display lighter or minimal hotspot intensity.

Overall, the heatmap indicates that urban centers and high-population districts experience substantially more crime compared to outer regions. These findings highlight the need for increased patrolling, stronger surveillance systems, and situational awareness in the highlighted zones.

iv. Visualization Interpretation

The heatmap uses color intensity to represent the severity of crime concentration:

- Red zones represent extreme hotspot locations with very high crime frequency.
- Orange and yellow zones indicate moderate crime density.
- Green or faint areas represent low or scattered incident zones.

The visualization provides an intuitive understanding of spatial crime distribution and helps prioritize areas requiring immediate safety interventions.



4.5 Victim Demographics

4.5.1 Victim Age Distribution

i. General Description

Age distribution provides insights into which age groups are most vulnerable to crime. This helps understand behavioral, social, or situational factors that influence victimization.

ii. Specific Requirements

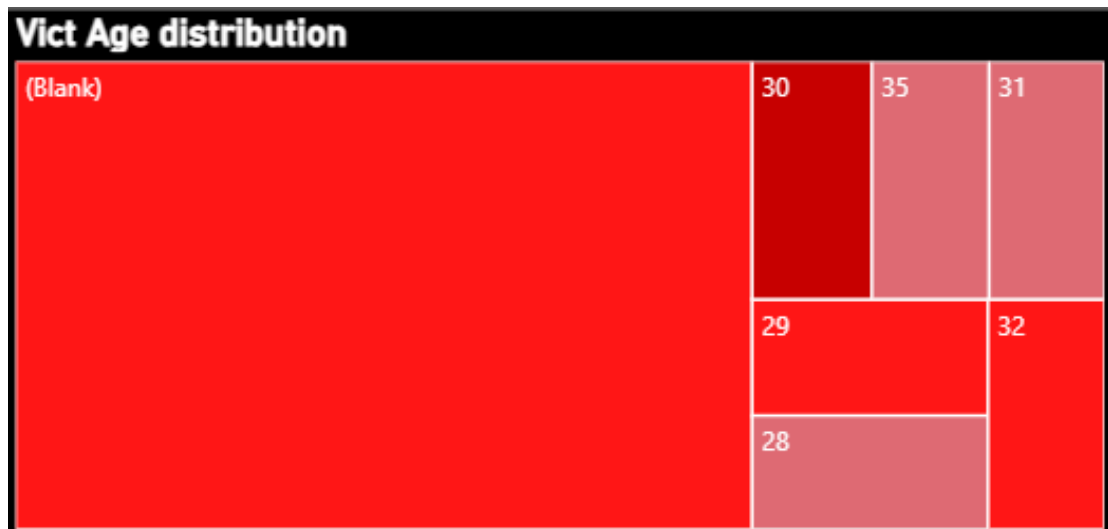
- Identify common age ranges of victims.
- Detect unusual patterns or missing values.
- Understand age-related risk exposure.

iii. Analysis Results

The treemap shows significant missing age values ("Blank"), followed by dominant victim ages between **28–35 years**. This indicates that young adults are the most frequent victims, possibly due to higher outdoor activity and workforce involvement.

iv. Visualization Interpretation

The chart reflects that most crime victims fall within the working-age population, highlighting increased exposure in public or transit environments.



4.5.2 Victim Descent

i. General Description

Victim descent analysis helps identify demographic groups disproportionately affected by crime. It provides important insights for community-specific interventions.

ii. Specific Requirements

- Calculate descent-wise distribution of victims.
- Compare proportions among different groups.

iii. Analysis Results

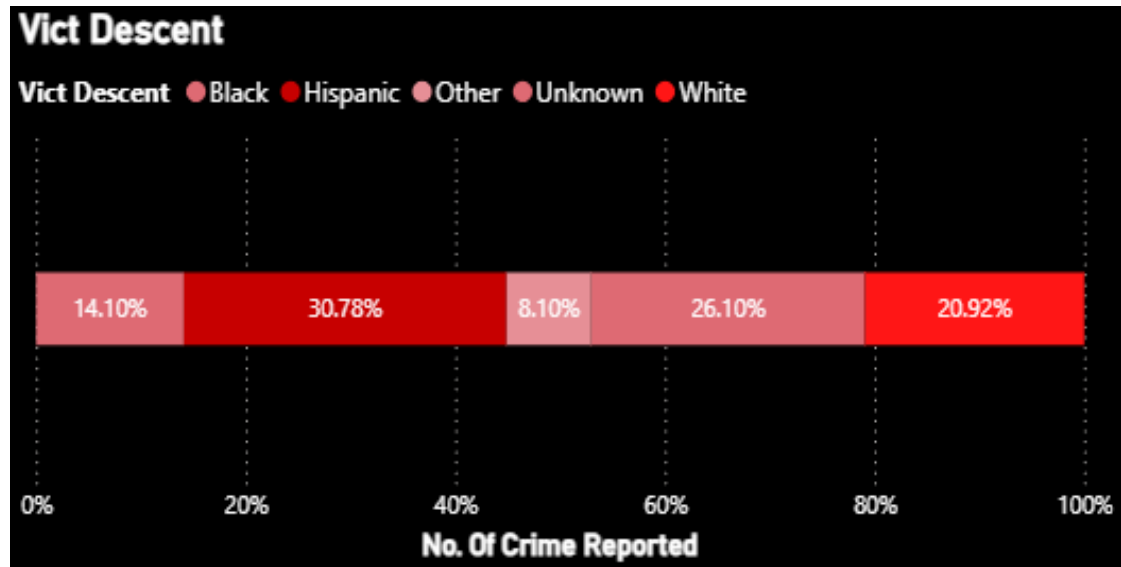
Largest affected groups:

- Hispanic – **30.78%**
- White – **20.92%**
- Unknown – **26.10%**
- Black – **14.10%**
- Other – **8.10%**

Hispanic victims form the largest group, followed by White and Unknown.

iv. Visualization Interpretation

The 100% stacked bar chart clearly reveals demographic disparities, indicating some communities may require focused safety strategies.



4.5.3 Victim Sex

i. General Description

Analyzing crime victims by sex helps identify gender-based crime vulnerability.

ii. Specific Requirements

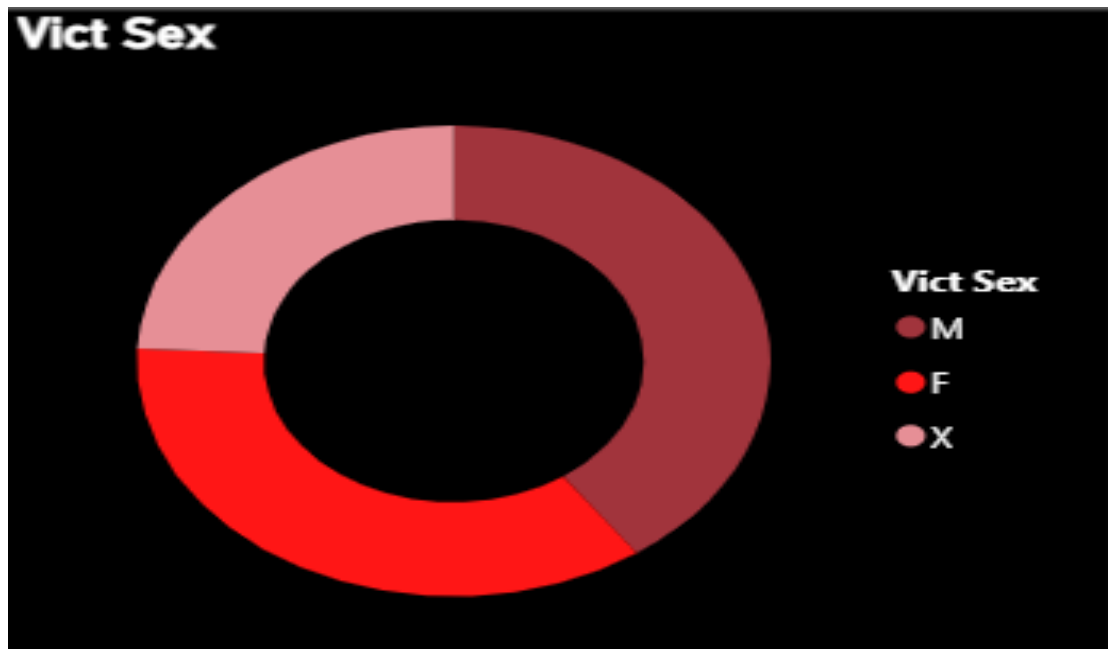
- Identify the distribution of victims across M, F, and X.
- Understand gender involvement in reported crimes.

iii. Analysis Results

Male victims form the largest portion, followed by Female victims. A small percentage belongs to the “X” category (non-binary/unspecified).

iv. Visualization Interpretation

The donut chart indicates male victims are more frequent, which may relate to higher public exposure or risk-related activities.



4.6 Weapon Usage Analysis

i. General Description

Weapon involvement significantly increases the severity and complexity of crime incidents. Understanding weapon usage trends helps assess threats and develop preventive measures.

ii. Specific Requirements

- Identify most commonly used weapons.
- Compare weapon categories by total crime involvement.
- Detect high-risk weapon types.

iii. Analysis Results

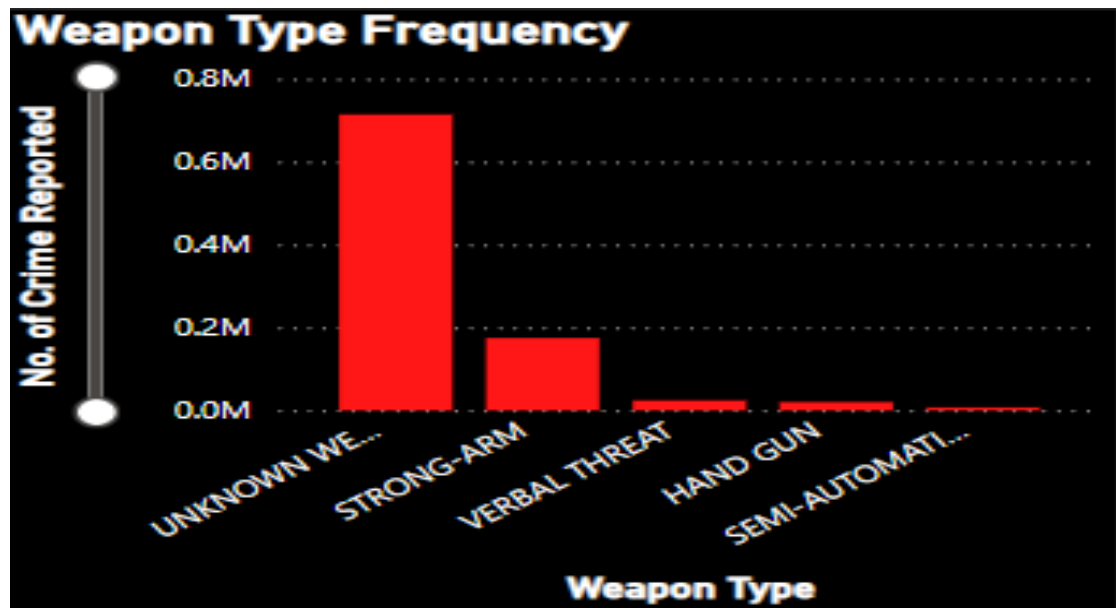
Most used weapon categories include:

1. **Unknown Weapon (largest category)**
2. **Strong-Arm (physical force)**
3. **Verbal Threat**
4. **Handgun**

Other firearms and weapons appear with smaller frequencies.

iv. Visualization Interpretation

Since “Unknown Weapon” is extremely high, many reports lack detailed weapon documentation. Strong-arm crimes suggest physical altercations, while handgun incidents reflect serious violent offenses.



4.7 Crime by Premise Type

i. General Description

Premise analysis identifies where crimes most frequently occur, helping in location-based policing.

ii. Specific Requirements

- Identify top premises where crimes occur.
- Compare crime levels across residential, street, and commercial areas.

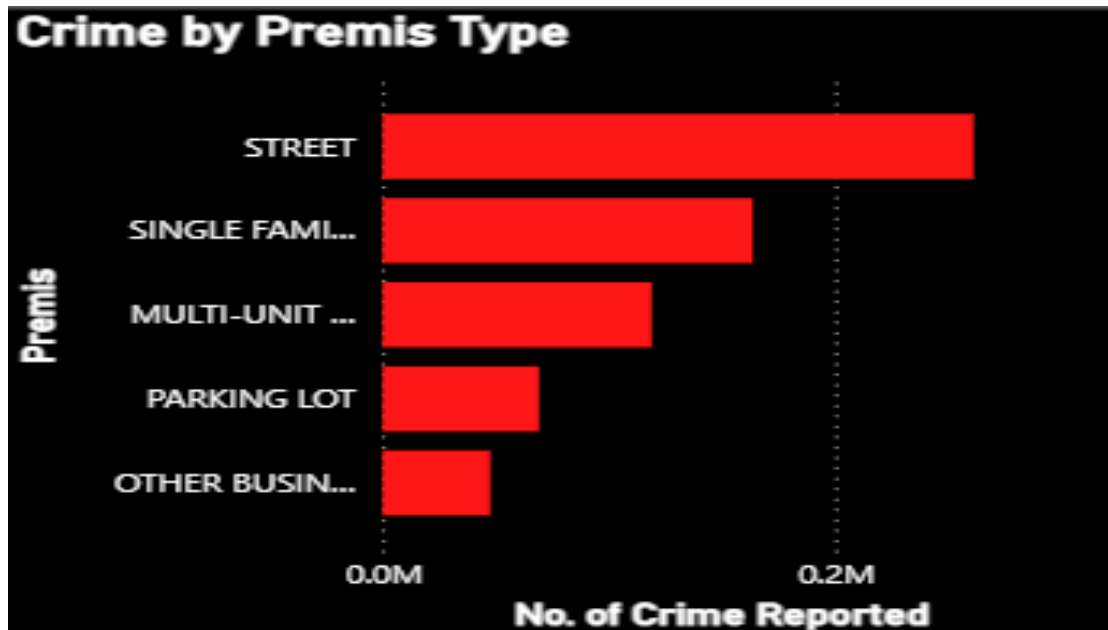
iii. Analysis Results

Top premises include:

- **Streets** (highest crime concentration)
- Single-Family Homes
- Multi-Unit Apartments
- Parking Lots
- Business Establishments

iv. Visualization Interpretation

“Street” as the highest category indicates outdoor exposure and public mobility increase crime likelihood.



4.8 Crime Hotspots by Area

i. General Description

Hotspot analysis identifies geographic zones with concentrated crime activity. These insights support targeted policing and resource allocation.

ii. Specific Requirements

- Rank areas by total reported crimes.
- Detect regions needing priority policing.

iii. Analysis Results

Top hotspot areas include:

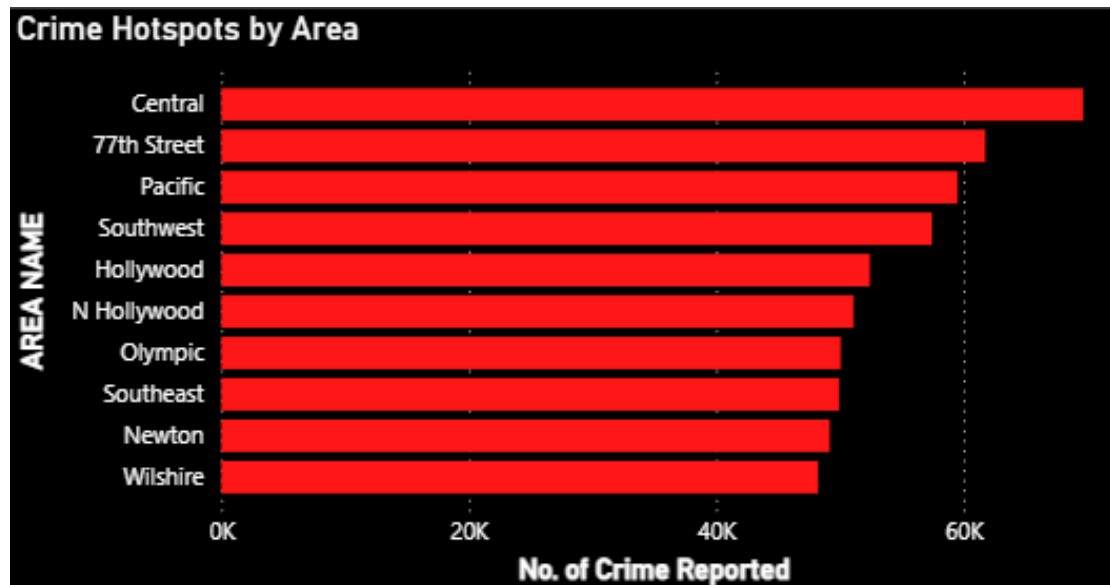
1. Central (highest)
2. 77th Street
3. Pacific
4. Southwest
5. Hollywood

6. North Hollywood

These areas consistently report higher crime activity.

iv. Visualization Interpretation

The bar chart highlights Central and 77th Street divisions as highly vulnerable, requiring focused patrol operations.



4.9 Case Status Breakdown

i. General Description

Case status analysis helps determine how many cases are resolved versus pending or uncertain.

ii. Specific Requirements

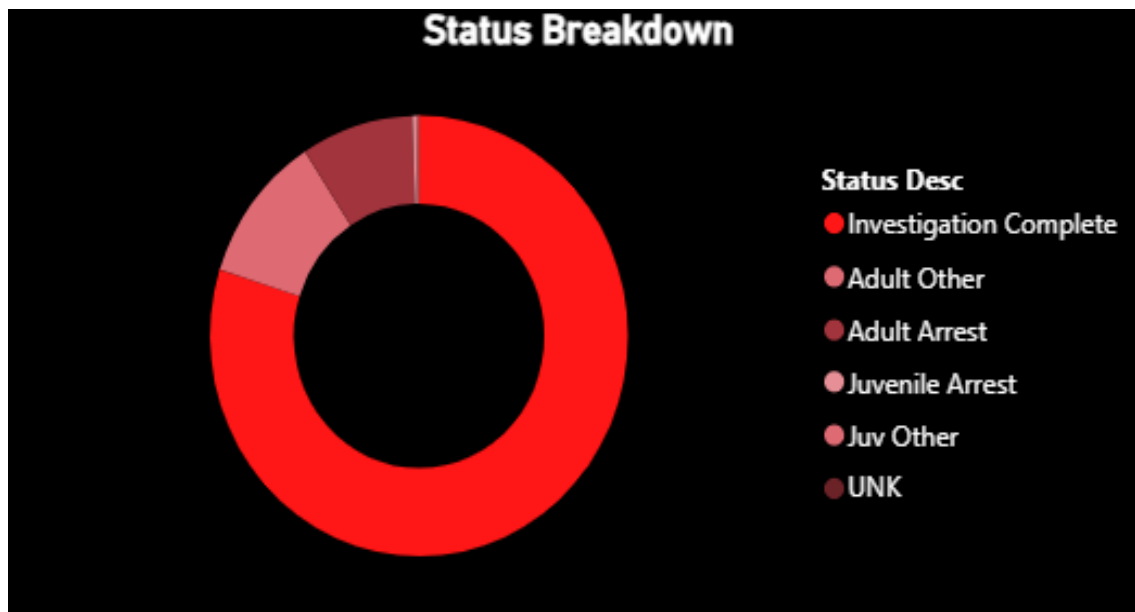
- Categorize cases by status.
- Identify proportion of completed vs ongoing investigations.

iii. Analysis Results

Most cases fall under **Investigation Complete**, followed by Adult Arrest and Juvenile Arrest. Small portions belong to "Other" and "Unknown".

iv. Visualization Interpretation

The donut chart emphasizes strong closure rates, suggesting efficient investigative follow-up for most cases.



4.10 District-Level Crime Analysis

i. General Description

District-level analysis enables comparison of crime frequency across police districts, helping identify high-risk operational zones.

ii. Specific Requirements

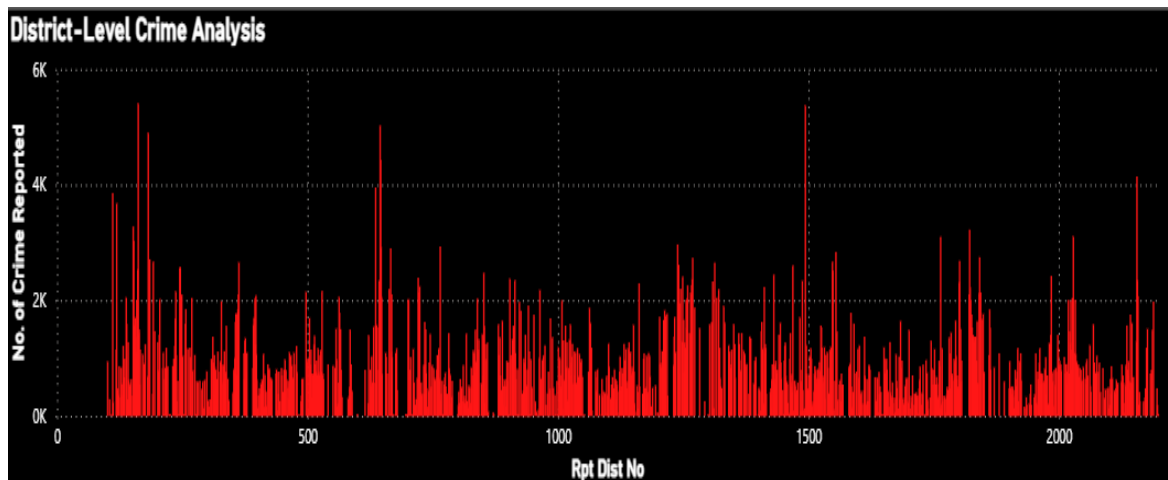
- Analyze crime count for all district numbers.
- Detect districts with unusual spikes.

iii. Analysis Results

Crime frequency varies significantly across districts, with some experiencing spikes exceeding **5,000** reported crimes. These peaks indicate operational hotspots needing enhanced monitoring.

iv. Visualization Interpretation

The line chart highlights fluctuating district crime levels, making it easy to locate districts with consistently high incident density.



5. Key Findings

1. **Crime peaked in 2023**, with the highest number of reported incidents, before showing a major decline in 2024 and 2025. This indicates a significant change in crime patterns or reporting behavior.
2. **Vehicle Theft is the most common crime type**, followed by Vandalism, Theft from Motor Vehicle, and Battery. Vehicle-related crimes dominate the dataset, highlighting mobility-related vulnerabilities.
3. **Afternoon and Evening are the peak crime hours**, accounting for the majority of incidents. This aligns with periods of high public activity and movement.
4. **Hotspot regions include Central, 77th Street, Pacific, Southwest, and Hollywood**, which consistently show high crime density. These areas require targeted patrol and surveillance.
5. **Victims are primarily young adults**, especially between ages **28–35**, making them the most vulnerable demographic group.
6. **Hispanic and White descent groups account for most victim records**, with a large “Unknown” category indicating incomplete reporting.
7. **Male victims are reported more frequently than females**, suggesting higher exposure to risk or involvement in public activities.
8. **Unknown Weapon and Strong-Arm Force are the leading weapon categories**, followed by Verbal Threat and Handgun.

This shows that many crimes involve physical force or lack detailed weapon documentation.

9. **Streets are the most common crime premises**, followed by residential and commercial areas. This highlights the need for improved street surveillance and community safety measures.
10. **Case Status analysis shows a strong percentage of completed investigations**, including Adult and Juvenile arrests, which reflects effective follow-up by law enforcement.

6. Conclusion

The *Crime Insights & Intelligence Portal* successfully transforms complex crime datasets into meaningful, interactive, and actionable intelligence. By leveraging Power BI for visual analytics, the project provides a comprehensive understanding of criminal activity across Los Angeles from 2020 to the present.

The analysis revealed several important findings:

- **Crime levels increased steadily until 2023** before significantly declining in 2024–2025, indicating shifts in reporting patterns or improved enforcement strategies.
- **Vehicle-related crimes** emerged as the most dominant crime category, highlighting the need for enhanced surveillance in public mobility zones and parking areas.
- **Afternoon and evening hours** were identified as the peak crime periods, suggesting that increased public movement and workplace transitions elevate risk.
- **Victim demographics** showed that young adults (28–35) and Hispanic communities were disproportionately affected, emphasizing demographic vulnerability.
- **Weapon usage analysis** revealed that strong-arm force and unknown weapon categories constitute the majority of reported cases, indicating a combination of physical altercations and incomplete reporting.

- **Hotspot analysis** identified Central LA, 77th Street, Pacific, Southwest, and Hollywood divisions as major crime epicenters, requiring targeted resource allocation.
- **Case status breakdown** showed a strong proportion of completed investigations and arrests, reflecting effective follow-up by law-enforcement units.
- **District-level analysis** allowed pinpointing the most volatile policing zones with crime spikes exceeding thousands of incidents.

Overall, the portal enhances situational awareness, supports evidence-based decision-making, and enables law-enforcement agencies to optimize patrol deployment, strengthen community safety strategies, and anticipate crime trends.

The project demonstrates the power of data analytics in criminology—turning raw data into insights that can significantly improve public safety and operational effectiveness.

7. Future Scope

While the *Crime Insights & Intelligence Portal* provides valuable insights into crime patterns, victim demographics, hotspot regions, and weapon usage, there are several opportunities to further enhance its analytical depth and operational effectiveness. Future improvements can significantly expand the system's impact and usability:

1. Integration of Predictive Analytics

Machine learning models such as time-series forecasting, hotspot prediction, and crime classification can be integrated to:

- Predict future crime trends
- Identify potential emerging hotspots
- Flag high-risk zones before incidents occur

This would transform the dashboard from a descriptive tool into a predictive intelligence system.

2. Real-Time Crime Monitoring

Connecting the portal to live police data feeds or IoT-based surveillance systems would enable:

- Real-time crime detection
- Instant hotspot updates
- Faster decision-making for ground units

This upgrade would support dynamic resource deployment.

3. Advanced Geospatial Analysis

Future versions may incorporate:

- Route-level danger scoring
- Street-segment crime forecasting
- Drone or GPS-integrated patrol suggestions

This will create a more granular and operationally actionable mapping experience.

4. Suspect Profiling & Behavioral Insights

By incorporating additional datasets (arrest records, previous offenders, gang activity logs), the portal could provide:

- Repeat offender tracking
- Criminal behavior patterns
- Risk scores for specific offender categories

This aids in targeted intervention and preventive policing.

5. Integration with Public Safety Systems

The portal can be extended to support:

- Emergency response automation
- Citizen reporting dashboards
- Collaboration with city departments for resource planning

Such integration would make the system part of a broader smart-city ecosystem.

6. Enhanced Data Quality & Enrichment

Future development can include:

- Automated data cleaning pipelines
- Natural Language Processing (NLP) to extract insights from police reports
- Cross-referencing with socioeconomic and traffic datasets

This would enable more comprehensive and context-rich analysis.

7. Interactive Storytelling Dashboards

Adding features like narrative insights, anomaly detection alerts, and AI-powered summaries will help:

- Non-technical users understand crime data easily
- Authorities receive automated briefings
- Enhance decision-making efficiency

8. Mobile & Web Deployment

Deploying the portal on mobile apps or responsive web platforms will enable:

- On-field officers to access insights instantly
- Decision-makers to monitor crime trends anytime, anywhere

This increases the system's accessibility and operational value.

8. References

- **Crime Dataset (2020–Present)**

Dataset Catalog. *Crime Data from 2020 to Present*.

(Dataset obtained from a public catalog for academic analysis, including

crime details, victim demographics, weapon usage, and geographic coordinates.)

- **Microsoft Power BI Documentation**

Microsoft Corporation. *Power BI Learning and Documentation*.

Available at: <https://learn.microsoft.com/power-bi/>

(Used for data modeling, report development, and visualization.)

DashBoard Link: <https://app.powerbi.com/groups/me/reports/09a3cec0-a365-4708-966d-6299b04e008a/ebe619d14144d8026dae?experience=power-bi>

LinkedIn: <https://www.linkedin.com/feed/update/urn:li:activity:7403083218998255616/>

Github: <https://github.com/Deepakjaat23/Crime-Insights-Intelligence-Portal>